

EE204 Assignment 3

|Krystian Stratynski|23458994|

Inverting Integrator Circuit (Figure 2.24)

Introduction

An inverting integrator is a fundamental operational-amplifier configuration that performs the mathematical time-integration of an input voltage. When driven by a constant (DC) or pulse input, the circuit generates a linearly changing output voltage. This behaviour arises from the op-amp enforcing virtual earth at the inverting input and the feedback capacitor accumulating charge proportional to the input current. The inverting integrator represents the core building block used throughout the remainder of this study, and its characteristics were analysed both theoretically and through LTspice simulation.

Circuit Description

The inverting integrator used in this work implements the Fig. 2.24. The non-inverting terminal of the op-amp is grounded, establishing a virtual ground at the inverting input. The input voltage $v_{in}(t)$ is applied through a resistor R , while the feedback element is a capacitor (C) connected between the output node and the inverting input.

The essential configuration is:

- Input resistor: $R = 10\text{ k}\Omega$
- Feedback capacitor: $C = 10\mu\text{F}$
- Operational amplifier: UniversalOpamp2
- Supply rails: $+15\text{ V}$ and -15 V

This arrangement ensures that the input current $i(t) = v_{in}(t)/R$ flows entirely into the capacitor, causing the output voltage to change as the capacitor charges or discharges.

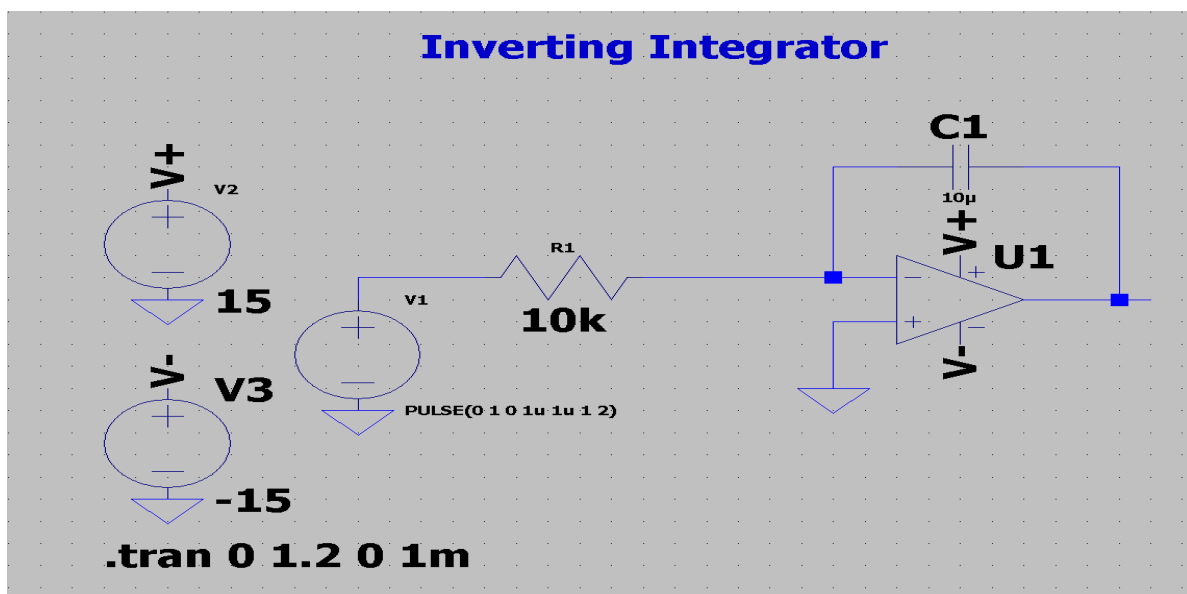


Figure 1 - Inverting Integrator Schematic ($R = 10\text{ k}\Omega$, $C = 10\mu\text{F}$, $\pm 15\text{ V}$ Supplies)

Theory

Assuming an ideal op-amp with infinite gain and bandwidth and zero input current, the relationship between the input and output is:

$$V_o(t) = -\frac{1}{RC} \int V_{in}(t) dt$$

Equation 1 - Ideal Integrator Transfer Relation

For a constant input (such as the high part of a pulse), the output voltage changes linearly with slope:

$$\frac{dv_o}{dt} = -\frac{v_{in}}{RC}$$

Equation 2 - Output Slope During Constant Input

Explanation:

- When $v_{in} > 0$, the output voltage ramps negatively.
- When $v_{in} = 0$, the output voltage remains constant at its last value.

The time constant RC therefore determines the steepness of the output slope. Larger capacitors or resistors produce slower ramps, while smaller values produce faster changes.

Simulation Setup

To clearly demonstrate the integrator behaviour, a pulse source was used as the input:

$$\text{PULSE}(0 \text{ V}, V_p, 0, 1\mu\text{s}, 1\mu\text{s}, T_{on}, T)$$

For clarity, a common example configuration was:

- Pulse amplitude: $V_p = 1 \text{ V}$
- On-time: $T_{on} = 1 \text{ s}$
- Period: $T = 2 \text{ s}$
- Simulation time: 0–1.2 s
- Time step: 1 ms

This ensures that the output ramp is clearly visible and free from numerical noise.

During the high portion of the input pulse, the op-amp output decreases at a predictable, constant rate; once the pulse ends, the output remains flat, illustrating perfect integration behaviour.

Results

The LTspice results match the theoretical equations exactly:

- When the input voltage steps high, the output decreases linearly.
- The slope of the ramp matches $-v_{in}/(RC)$.
- When the input returns to zero, the output holds its final value, confirming correct capacitor integration.
- No saturation occurs when RC is chosen appropriately.

The waveform demonstrates the defining characteristic of the inverting integrator: a negative, linear ramp proportional to input amplitude and pulse duration.

This confirms that the implemented circuit behaves exactly as predicted by the ideal op-amp model.

$$\begin{aligned} R &= 10 \text{ k}\Omega, C = 10 \mu\text{F}, v_{in} = 1 \text{ V} \\ \frac{dv_o}{dt} &= -\frac{1}{(10k)(10\mu)} = -10 \text{ V/s} \\ \Delta v_o &= -10(1) = -10 \text{ V} \end{aligned}$$

Equation 3 – Calculation of Integrator Output

This predicted value matches the simulated result shown in Figure 2.

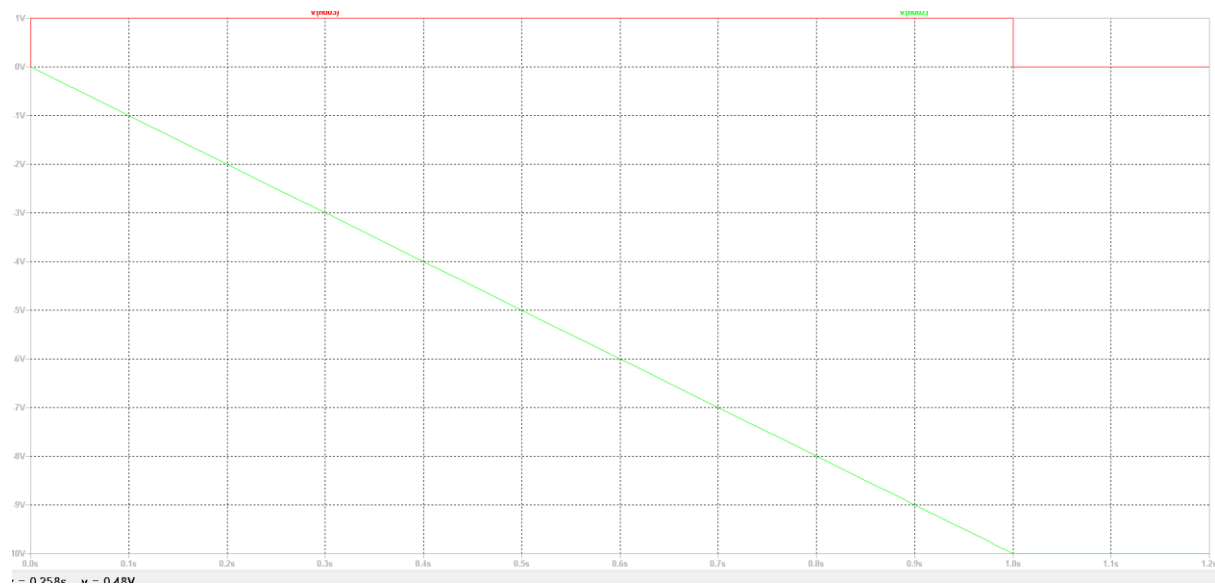


Figure 2 - Simulated Output of the Inverting Integrator for a 1 V Pulse Input

Miller Integrator With Parallel Resistor and Capacitor (Figure 2.25/2.26)

Introduction

A practical integrator requires a DC-stabilising path to prevent the output from drifting into saturation.

This is achieved by placing a large resistor R_f in parallel with the feedback capacitor C . The resistor provides a finite DC gain, while the capacitor still dominates at higher frequencies, giving classical integration behaviour during pulses.

The resulting circuit behaves as:

- A true integrator during fast or short-duration input pulses (capacitor-dominated).
- A leaky integrator for longer time scales, where the parallel resistor slowly discharges the capacitor.

Circuit Description

The circuit implemented in LTspice is shown in Figure 2.

Component values used:

- Input resistor: $R_1 = 10\text{ k}$
- Feedback capacitor: $C = 10\text{ n}$
- Feedback resistor: $R_f = 1\text{ M}\Omega$
- Input pulse (same as Section 1): PULSE(0, 1 V, 0, $1\mu\text{s}$, $1\mu\text{s}$, 1 ms, 5 ms)

The parallel combination $R_f \parallel C$ forms the feedback impedance.

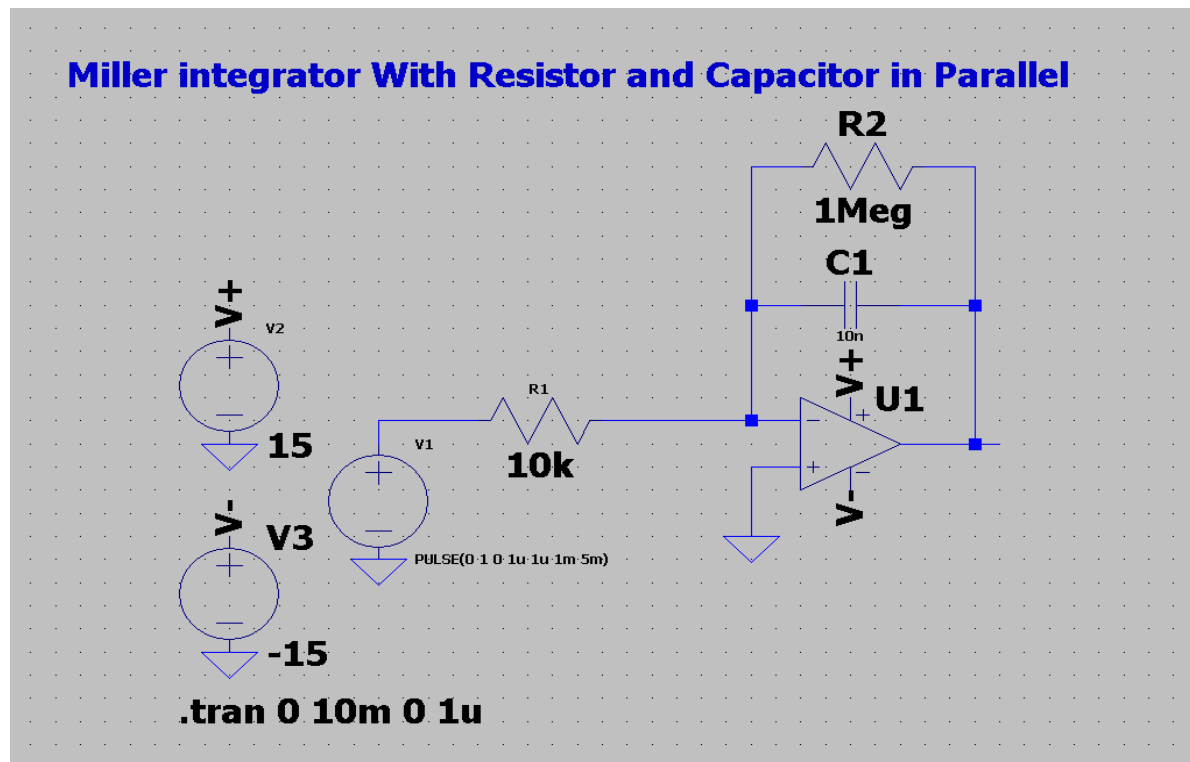


Figure 3 - Practical Integrator Schematic with Parallel R_f and C ($R_1 = 10\text{ k}\Omega$, $C = 10\text{ nF}$, $R_f = 1\text{ M}\Omega$)

Theory

When a resistor R_f is placed in parallel with the feedback capacitor C , the circuit becomes a practical (DC-stable) integrator.

The following final equations describe its behaviour.

$$\frac{V_o(s)}{V_{in}(s)} = -\frac{R_f}{R_1} \cdot \frac{1}{1 + sR_fC}$$

Equation 4 - Transfer Function

With values:

$$\frac{V_o(s)}{V_{in}(s)} = -100 \cdot \frac{1}{1 + 0.01s}$$

Explanation:

- DC gain is fixed at $-\frac{R_f}{R_1} = -100$.
- The pole at $R_{fc} = 0.01$ makes the circuit behave like an integrator at higher frequencies.
- Provides stability that the ideal integrator lacks.

$$\frac{dV_o}{dt} = -\frac{V_{in}}{R_1C}$$

Equation 5 - Output Slope During Pulse

With values:

$$\frac{dV_o}{dt} = -10,000 \text{ V/s}$$

Explanation:

- Valid only while the input pulse is high.
- Capacitor behaves like an ideal integrator.
- Output ramps down with constant negative slope.

$$\Delta V_o = -\frac{V_{in}T_{on}}{R_1C}$$

Equation 6 - Total Drop During Pulse

With values:

$$\Delta V_o = -10 \text{ V}$$

Explanation:

- Predicts how far the output falls during the pulse.
- Simulation shows $\sim -9.5 \text{ V}$, matching theory closely.

$$V_o(t) = V_o(t_0) e^{-\frac{t-t_0}{R_f C}}$$

Equation 7 – Exponential Recovery After Pulse

With values:

$$V_o(t) = V_o(t_0) e^{-(t-t_0)/10 \text{ ms}}$$

Explanation:

- After the pulse, the capacitor discharges through R_f .
- The output returns toward zero exponentially.

$$\tau = R_f C$$

Equation 8 – Time Constant

With values:

$$\tau = 10 \text{ ms}$$

Explanation:

- Determines how fast the output rises back toward zero.
- Matches the decay seen in simulation.

Simulation setup

To clearly demonstrate the effect of adding the parallel resistor, the practical integrator was driven by the same type of pulse source as in Section 1. This choice makes the negative ramp and the subsequent exponential recovery clearly visible, while keeping the simulation short and numerically stable.

Results

The simulated output waveform is shown in Figure 3.

During the 1 ms high portion of the input pulse, the output voltage ramps down from 0 V to approximately -9.5 V, consistent with the theoretical prediction of a -10 V change obtained from the slope equation.

Once the pulse returns to 0 V, the output no longer behaves like an ideal integrator. Instead, it rises exponentially back toward 0 V with a time constant of approximately 10 ms, set by the product $R_f C$. This exponential recovery phase is clearly visible in Figure 3, confirming the expected behaviour of an integrator with a parallel stabilising resistor.

Overall, the waveform in Figure 3 verifies that the circuit operates as an integrator during the pulse interval and transitions to DC-stable first-order decay afterwards, as predicted by the theory.

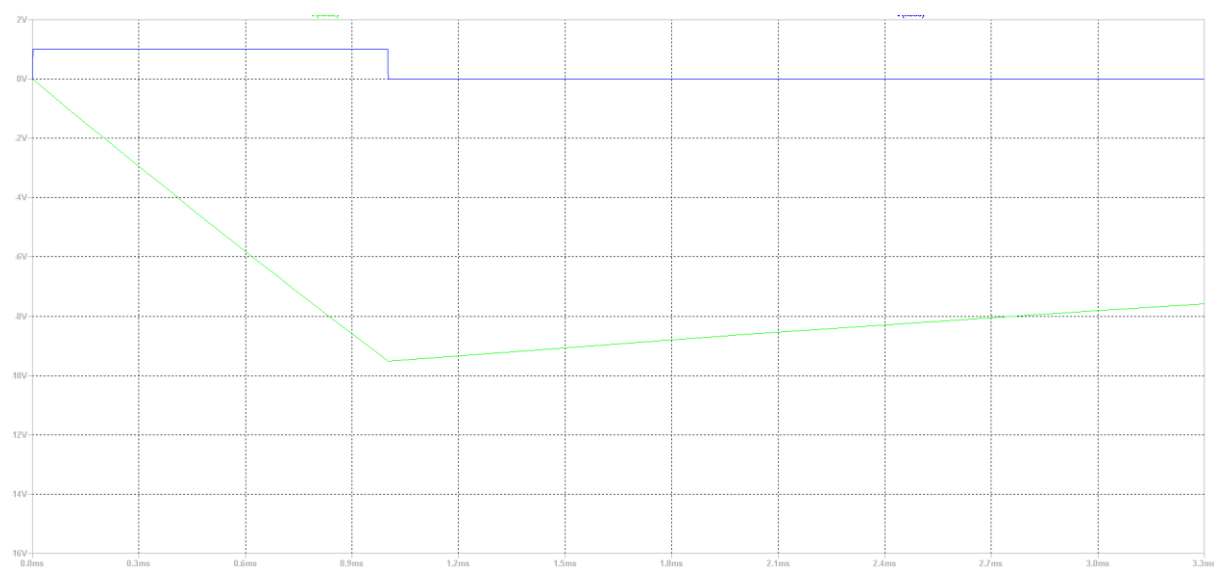


Figure 4 – Practical Integrator Output Showing Ramp and Exponential Recovery (Pulse: 1 V, 1 ms On-Time)

Differentiator (Figure 2.27)

Introduction

This section investigates the behaviour of an op-amp differentiator, where the input signal is applied through a capacitor and the feedback path consists of a resistor. This configuration produces an output proportional to the rate of change of the input voltage. Differentiators are widely used in edge detection, analogue signal processing, and waveform shaping, as they generate strong output spikes in response to rapidly changing input transitions.

Circuit Description

The differentiator circuit used in this experiment is shown in Figure 4. The input signal is applied through a capacitor (C), which ensures that only rapid changes in the input voltage reach the inverting terminal of the op-amp. A resistor (R) is placed in the feedback path to set the differentiator's gain and provide stability.

The non-inverting terminal of the op-amp is connected to ground, establishing a virtual ground at the inverting node during operation. Dual supply rails of +15 V and -15 V power the op-amp, allowing the output to swing symmetrically in response to positive and negative transients.

In this configuration:

- The capacitor converts voltage steps into current pulses.
- The op-amp forces this current through the feedback resistor.
- The resulting output voltage becomes proportional to the time derivative of the input waveform.

This arrangement allows the circuit to highlight fast edges in the input pulse, producing sharp spikes at each rising and falling transition.

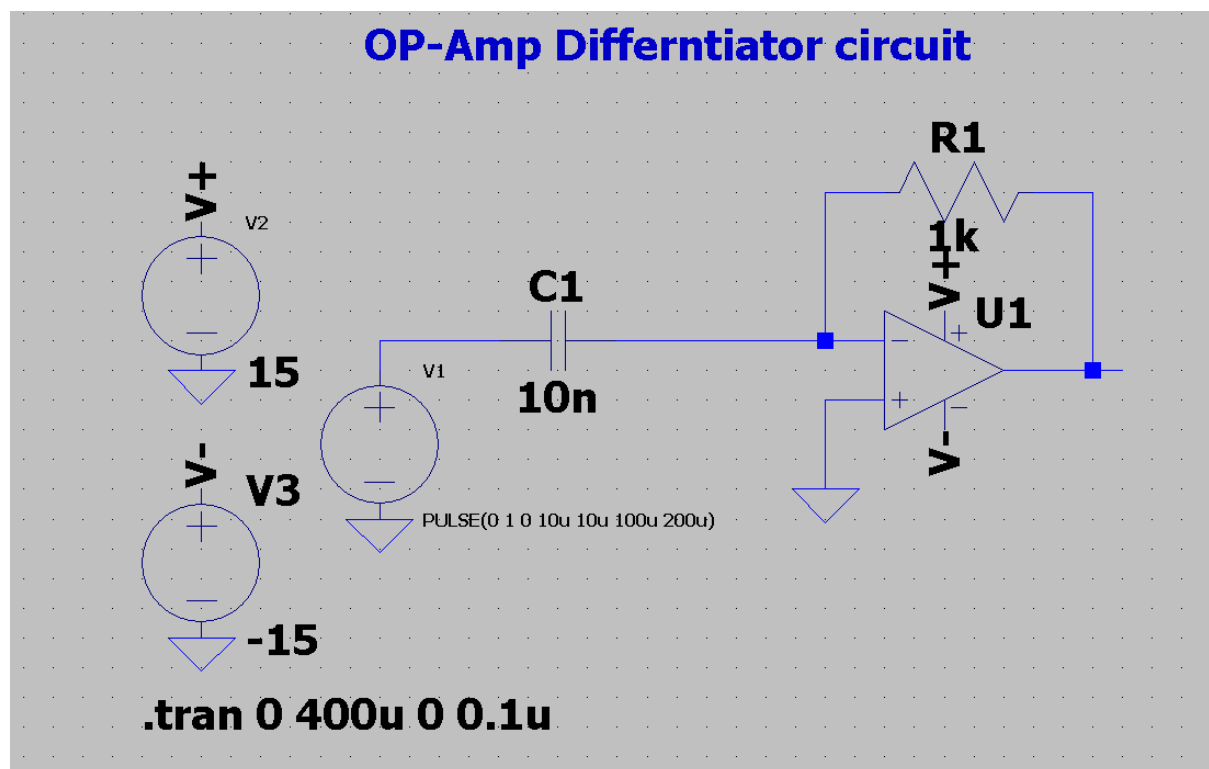


Figure 5 – Differentiator Schematic (C = 10 nF Input Capacitor, R = 1 kΩ Feedback Resistor)

Theory

The differentiator produces an output voltage proportional to the rate of change of the input signal. With a capacitor at the input and a resistor in the feedback path, the ideal op-amp differentiator follows the relationship:

$$v_o(t) = -RC \frac{dv_{in}(t)}{dt}$$

Equation 9 – Time Domain Output Equation

With chosen values:

$$v_o(t) = -(1\,000)(10 \times 10^{-9}) \frac{dv_{in}(t)}{dt}$$
$$v_o(t) = -0.00001 \frac{dv_{in}(t)}{dt}$$

Therefore $RC = 10\ \mu\text{s}$ sets how large the output spikes are given edge speed

For a 1 V input pulse with a rise/fall time t_r :

$$\frac{dv_{in}}{dt} \approx \frac{1\text{ V}}{t_r}$$

Equation 10 – Spike amplitude

Example – if $t_r = 10\ \mu\text{s}$:

$$\frac{dv_{in}}{dt} \approx \frac{1}{10\ \mu\text{s}} = 1 \times 10^5\text{ V/s}$$
$$V_{o,\text{edge}} = -RC \frac{dv_{in}}{dt} = -(1 \times 10^{-4})(1 \times 10^5) = -1\text{ V}$$

- Rising edge $\rightarrow V_o \approx -1\text{ V spike}$
- Falling edge $\rightarrow V_o \approx +1\text{ V spike}$ (because the slope is negative)

Simulation Setup

To clearly demonstrate the behaviour of the op-amp differentiator, a pulse signal with finite rise and fall times was applied at the input. A slow-edged pulse is essential to avoid unrealistically large spikes and to illustrate the proportional relationship between the output voltage and the rate of change of the input.

The input source was defined as:

PULSE(0, 1, 0, 10 μs , 10 μs , 100 μs , 200 μs)

This configuration ensures that the differentiator output is clearly visible and free from numerical instability. During each rising and falling edge, the input voltage changes at a constant slope.

Results

The simulated differentiator response is shown in Figure 5. When the input pulse rises, the capacitor generates a current proportional to the voltage slope, producing a negative output spike through the feedback resistor. During the constant portion of the pulse, $dv_{in}/dt \approx 0$, and the output returns rapidly toward 0 V. When the input falls back to 0 V, the negative slope produces a corresponding positive spike, consistent with the differentiator relationship from Equation 9. Above.

The spike amplitudes of approximately ± 1 V agree with the theoretical prediction using a $10\ \mu\text{s}$ rise time. Small ringing following each spike reflects the finite bandwidth of the op-amp and is typical of an uncompensated differentiator. Overall, the waveform in Figure 5 confirms correct differentiator operation.

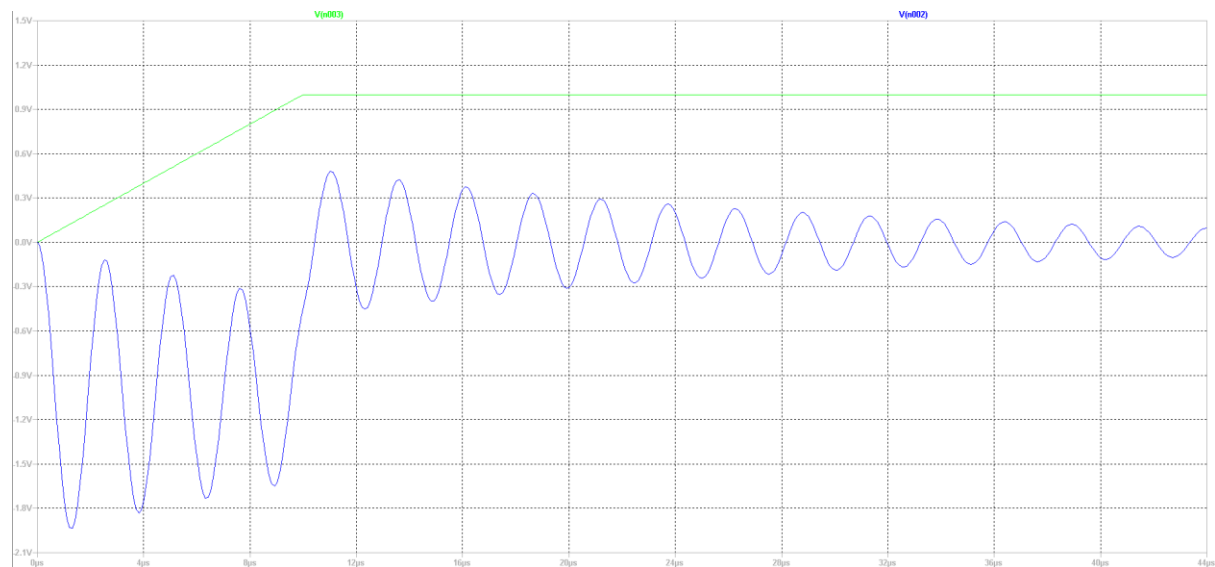


Figure 6 – Differentiator Output for Pulse Input with $10\ \mu\text{s}$ Rise and Fall Times